

CSE-421

Computer Graphics: Homogeneous Coordinates

Lecture-02

Lecturer, Varendra University

Why we need Homogeneous Coordinates?

- ◆ One of the many purposes of using homogeneous coordinates is to capture the concept of infinity.
- ◆ If we don't use homogeneous coordinates, it would be difficult to design certain classes of very useful curves and surfaces. These curves and surfaces are very crucial in developing algorithms in computer vision, graphics, CAD, etc.
- ◆ We have seen that basic transformations can be expressed in matrix form. But many graphic application involve sequences of geometric transformations. Hence we need a general form of matrix to represent such transformations.

Homogeneous coordinates

- Introduced in mathematics:
 - for projections and drawings
 - used in artillery, architecture
 - used to be classified material (in the 1850s)

- Add a third coordinate, w

- A 2D point is a 3 coordinates vector:

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

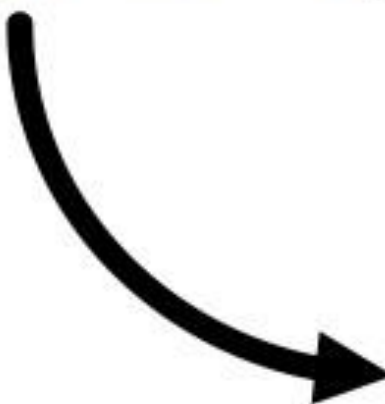
Homogeneous coordinates

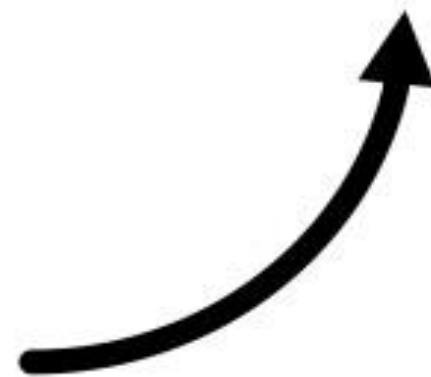
- Two points are equal if and only if:
 $x'/w' = x/w$ and $y'/w' = y/w$
- $w=0$: points at infinity
 - useful for projections and curve drawing
- Homogenize = divide by w .
- Homogenized points: $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

Translations with homogeneous

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

$$\begin{cases} \frac{x'}{w'} = \frac{x}{w} + t_x \\ \frac{y'}{w'} = \frac{y}{w} + t_y \end{cases}$$


$$\begin{cases} x' = x + wt_x \\ y' = y + wt_y \\ w' = w \end{cases}$$



Scaling with homogeneous

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

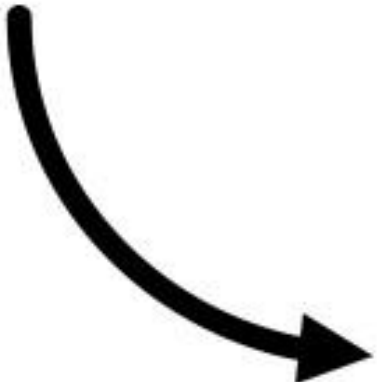
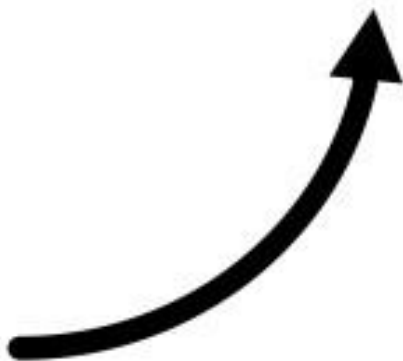
$$\begin{cases} \frac{x'}{w'} = s_x \frac{x}{w} \\ \frac{y'}{w'} = s_y \frac{y}{w} \end{cases}$$


$$\begin{cases} x' = s_x x \\ y' = s_y y \\ w' = w \end{cases}$$



Rotation with homogeneous

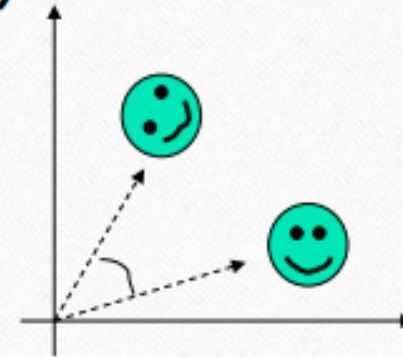
$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix} \quad \begin{cases} \frac{x'}{w'} = \cos \theta \frac{x}{w} - \sin \theta \frac{y}{w} \\ \frac{y'}{w'} = \sin \theta \frac{x}{w} + \cos \theta \frac{y}{w} \end{cases}$$


$$\begin{cases} x' = \cos \theta x - \sin \theta y \\ y' = \sin \theta x + \cos \theta y \\ w' = w \end{cases}$$


Rotation Revisit

- The standard rotation matrix is used to rotate about the origin (0,0)

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

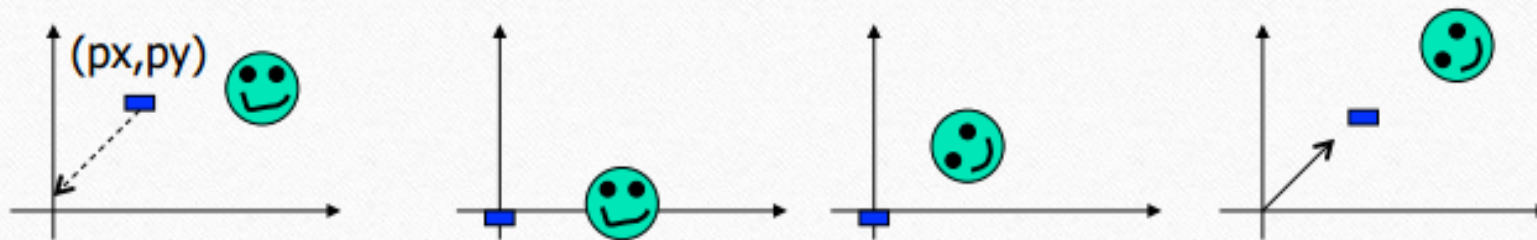


- What if I want to rotate about an arbitrary center?



Arbitrary Rotation Center

- To rotate about an arbitrary point $P (p_x, p_y)$ by θ :
 - Translate the object so that P will coincide with the origin: $T(-p_x, -p_y)$
 - Rotate the object: $R(\theta)$
 - Translate the object back: $T(p_x, p_y)$



Arbitrary Rotation Center

- Translate the object so that P will coincide with the origin: $T(-px, -py)$
- Rotate the object: $R(\theta)$
- Translate the object back: $T(px, py)$

- Put in matrix form: $T(px, py) R(\theta) T(-px, -py) * P$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & px \\ 0 & 1 & py \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -px \\ 0 & 1 & -py \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Composite Transformations

(A) Translations

If two successive translation vectors (t_{x1}, t_{y1}) and (t_{x2}, t_{y2}) are applied to a coordinate position P , the final transformed location P' is calculated as: -

$$\begin{aligned} P' &= T(t_{x2}, t_{y2}) \cdot \{T(t_{x1}, t_{y1}) \cdot P\} \\ &= \{T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1})\} \cdot P \end{aligned}$$

Where P and P' are represented as homogeneous-coordinate column vectors. We can verify this result by calculating the matrix product for the two associative groupings. Also, the composite transformation matrix for this sequence of transformations is: -

$$\begin{vmatrix} 1 & 0 & t_{x2} \\ 0 & 1 & t_{y2} \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 & t_{x1} \\ 0 & 1 & t_{y1} \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & t_{x1}+t_{x2} \\ 0 & 1 & t_{y1}+t_{y2} \\ 0 & 0 & 1 \end{vmatrix}$$

Or,
$$T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1}) = T(t_{x1}+t_{x2}, t_{y1}+t_{y2})$$

Which demonstrate that two successive translations are additive.

(B) Rotations

Two successive rotations applied to point P produce the transformed position: -

$$\begin{aligned} P' &= R(\Theta_2) \cdot \{R(\Theta_1) \cdot P\} \\ &= \{R(\Theta_2) \cdot R(\Theta_1)\} \cdot P \end{aligned}$$

By multiplication the two rotation matrices, we can verify that two successive rotations are additive:

$$R(\Theta_2) \cdot R(\Theta_1) = R(\Theta_1 + \Theta_2)$$

So that the final rotated coordinates can be calculated with the composite rotation matrix as: -

$$P' = R(\Theta_1 + \Theta_2) \cdot P$$

(C) Scaling

Concatenating transformation matrices for two successive scaling operations produces the following composite scaling matrix: -

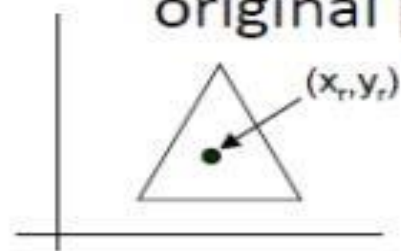
$$\begin{vmatrix} S_{x2} & 0 & 0 \\ 0 & S_{y2} & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} S_{x1} & 0 & 0 \\ 0 & S_{y1} & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} S_{x1} \cdot S_{x2} & 0 & 0 \\ 0 & S_{y1} \cdot S_{y2} & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Or,
$$S(S_{x2}, S_{y2}) \cdot S(S_{x1}, S_{y1}) = S(S_{x1} \cdot S_{x2}, S_{y1} \cdot S_{y2})$$

The resulting matrix in this case indicates that successive scaling operations are multiplicative.

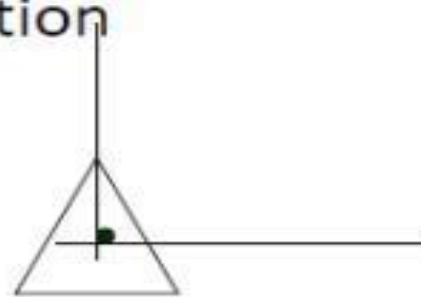
General pivot point rotation

- Translate the object so that pivot-position is moved to the coordinate origin
- Rotate the object about the coordinate origin
- Translate the object so that the pivot point is returned to its original position



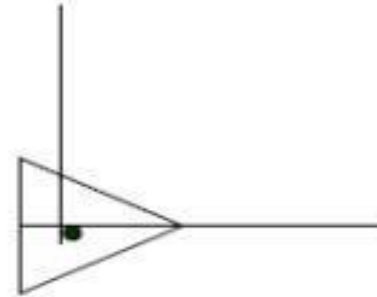
(a)

Original Position
of Object and
pivot point



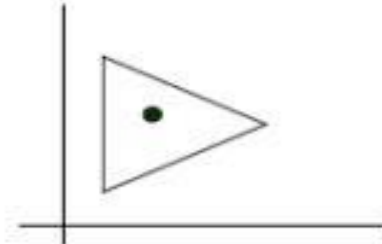
(b)

Translation of
object so that
pivot point (x_r, y_r)
is at origin



(c)

Rotation was
about origin

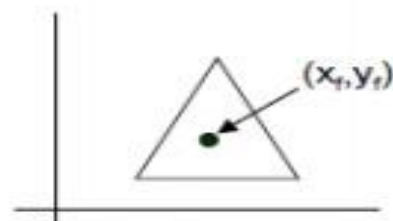


(d)

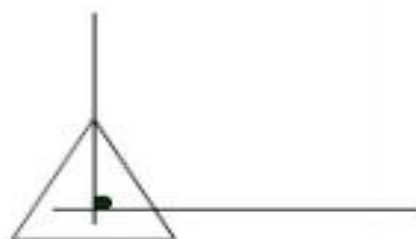
Translation of the object
so that the pivot point is
returned to position
 (x_r, y_r)

General fixed point scaling

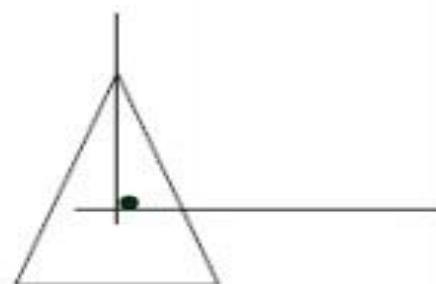
- Translate object so that the fixed point coincides with the coordinate origin
- Scale the object with respect to the coordinate origin
- Use the inverse translation of step 1 to return the object to its original position



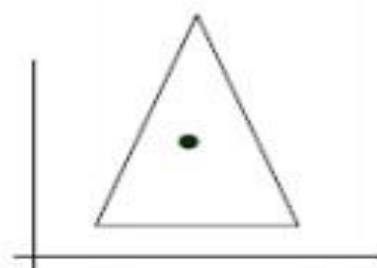
(a)
Original Position
of Object and
Fixed point



(b)
Translation of
object so that
fixed point
 (x_f, y_f) is at origin



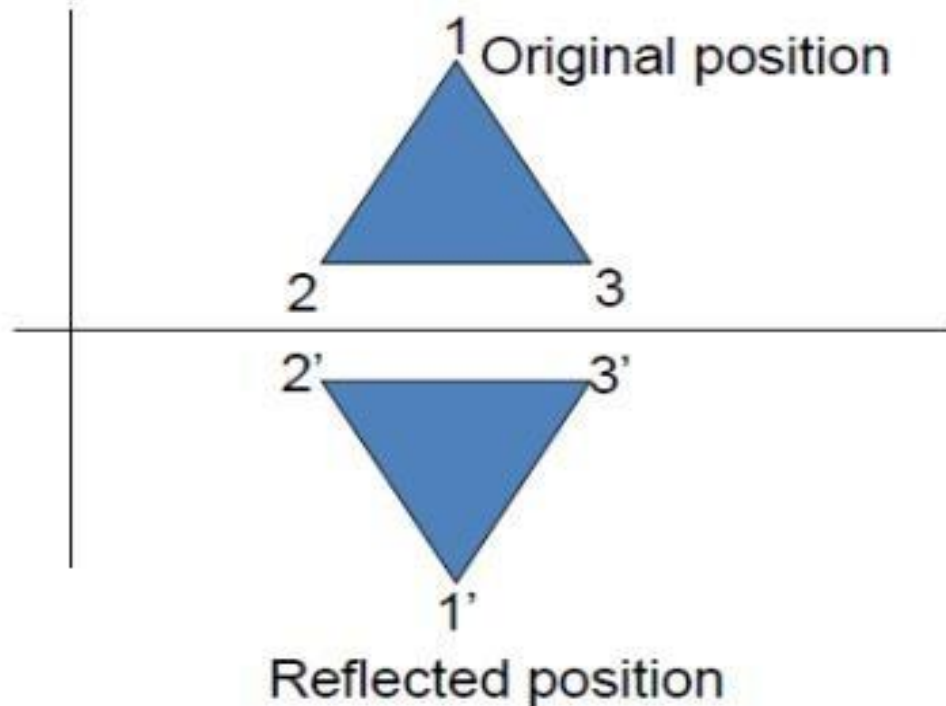
(c)
scaling was
about origin



(d)
Translation of the object
so that the Fixed point
is returned to position
 (x_f, y_f)

Other transformations

Reflection is a transformation that produces a mirror image of an object. It is obtained by rotating the object by 180 deg about the reflection axis

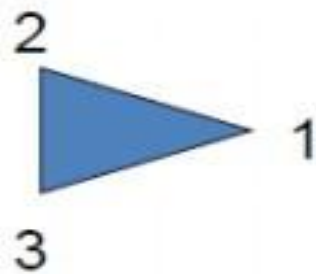


Reflection about the line $y=0$, the X- axis , is accomplished with the transformation matrix

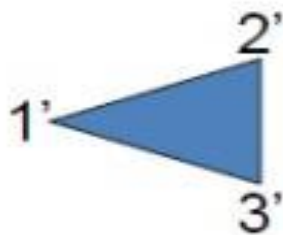
$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Reflection

Original position



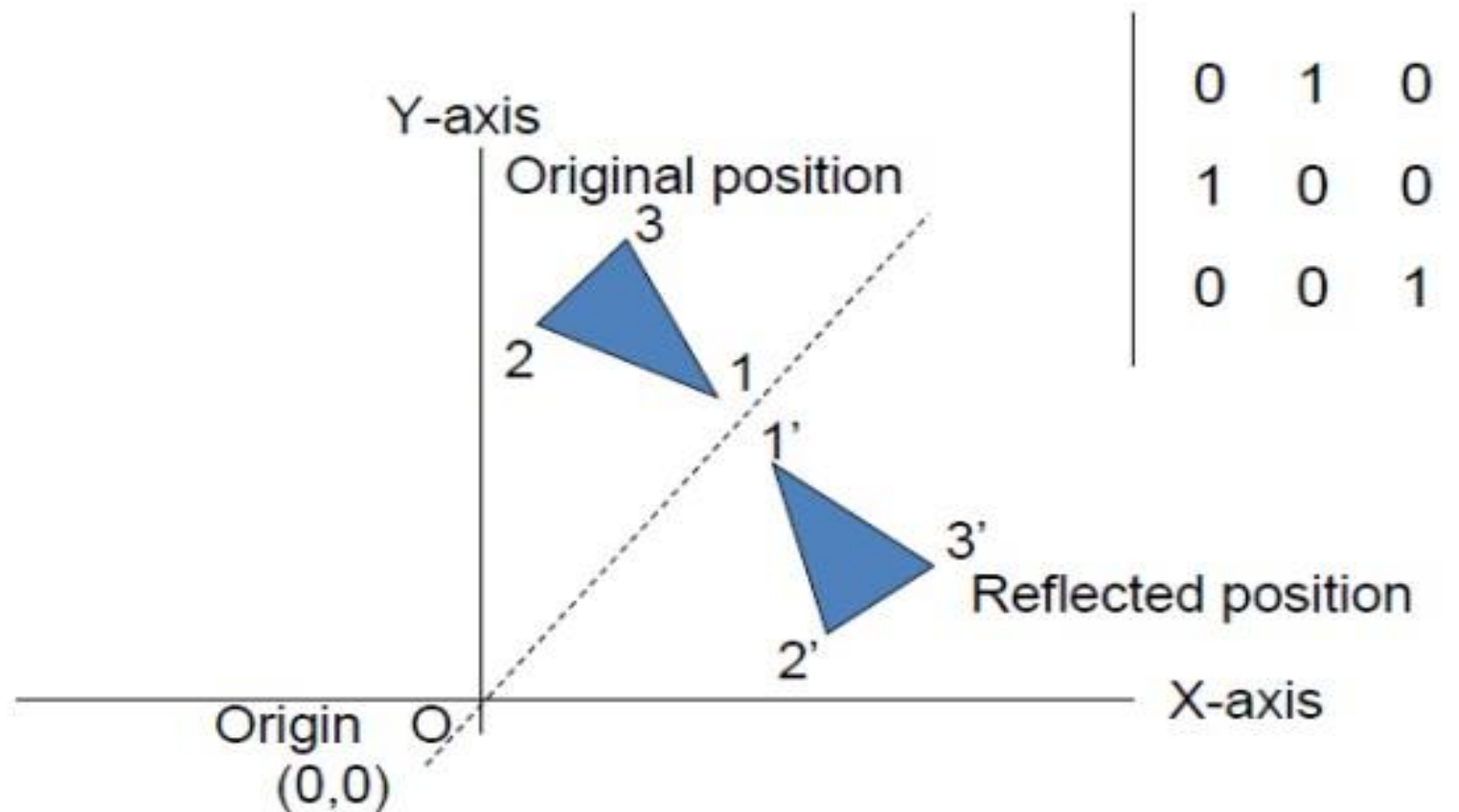
Reflected position



Reflection about the line $x=0$, the Y- axis , is accomplished with the transformation matrix

$$\begin{vmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Reflection of an object w.r.t the straight line $y=x$



Reflection About an Arbitrary Line:

1. Translate line to the Origin.
2. Rotation about the Origin.
3. Reflection Matrix.
4. Reverse the rotation.
5. Translate line Back.

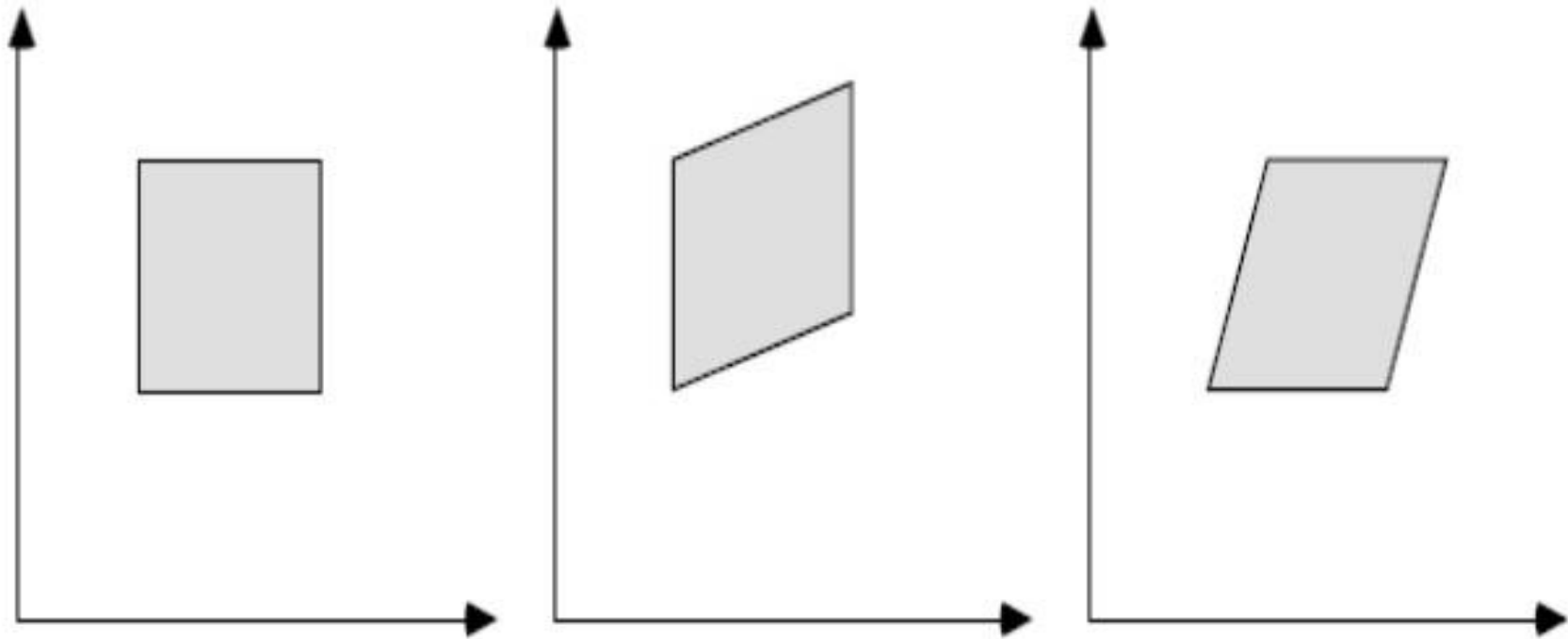
$$T(0, b) \cdot R_{\theta} \cdot R_{f_x} \cdot R_{-\theta} \cdot T(0, -b) \quad [\text{when, } y=mx+b]$$

Note: If the line passes the origin we need only 3 Steps (remove step 1 and step 5).

Shear Transformations

- Shear is a transformation that distorts the shape of an object such that the transformed shape appears as if the object were composed of internal layers that had been caused to slide over each other
- Two common shearing transformations are those that shift coordinate x values and those that shift y values

Shears



Original Data

y Shear

x Shear

Shear Transformation:

A transformation that slants the shape of an object.

Shear along the x-axis

$$SH_x = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} x+ay \\ y \\ 1 \end{bmatrix}$$

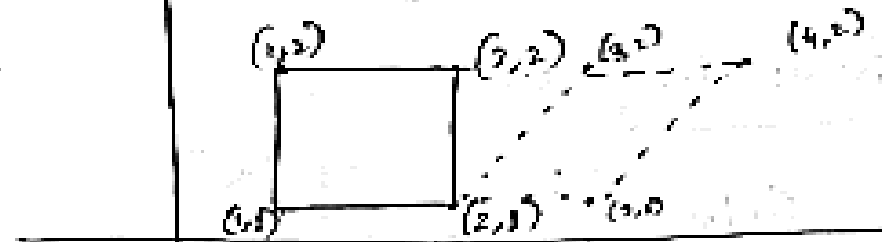
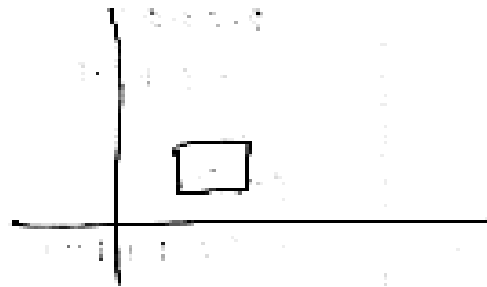
$$x' = x + ay$$

$$y' = y$$

Shear along the y-axis

$$SH_y = \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} x \\ y+bx \\ 1 \end{bmatrix}$$



$$a=1$$

$$x'_1 = x + ay$$

$$= 1 + 1 \cdot 1$$

$$= 2$$

$$(2, 1)$$

$$x'_3 = 1 + 1 \times 2$$

$$= 3$$

$$(3, 2)$$

$$x'_2 = 2 + 1 \times 1$$

$$= 2 + 1$$

$$= 3$$

$$(3, 1)$$

$$x'_4 = 2 + 1 \times 2$$

$$= 4$$

$$(4, 2)$$

- EX: Given 4 homogeneous points
- $P_0=(3,4,2,.5)$ $P_1=(24,32,16,4)$ $P_2=(9,12,6,1)$ $P_3=(18,24,12,3)$
- Which of the homogeneous points represent a different 3d point?
- $P_0 = (3/.5, 4/.5, 2/.5) = (6, 8, 4)$
- $P_1 = (24/4, 32/4, 16/4) = (6, 8, 4)$
- $P_2 = (9/1, 12/1, 6/1) = (9, 12, 6)$
- $P_3 = (18/3, 24/3, 12/3) = (6, 8, 4)$
- So p_2 is different

- EX: Apply two scaling to the matrix such that $S_{x1}=2$, $S_{y1}=3$, $S_{x2}=4$, $S_{y2}=2$.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Way #1

$$\bullet P' = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\bullet P'' = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Way #2

- $$T = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- $$P'' = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- EX: Suppose you have the point (2,4), rotate the point by 90 degree then by 180 degree then by 45 degree then by 45 degree then scale it by $S_x=2$ and $S_y=4$.
- Composite rotation : $90+180+45+45 = 360$ degree

Now solve it.....

COMMUTIVITY OF TRANSFORMATIONS

- Cases where $T1 * T2 = T2 * T1$:

T1	T2
Translation	Translation
Scaling	Scaling
Rotation	Rotation
Uniform Scaling	Rotation

- EX: If we scale, translate to origin then translate back is it equivalent to translate to origin, scale then translate back? ($S_x=2$, $S_y=3$, $t_x=3$, $t_y=2$)
- Answer: No, the order of matrix is important.

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 3 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

- EX: If we rotate the object by 90 degree then scale it ,is it equivalent to scale it then rotate it by 90 degree ($S_x=2, S_y=3$)

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -3 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -2 & 0 \\ 3 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **EX:** If we rotate the object by 90 degree then scale it ,is it equivalent to scale it then rotate it by 90 degree ($S_x=4, S_y=4$)

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Composition of 2D Transformation

Example: Find transformation of a triangle
With A(1,0), B(0,1) and C(1,1) by

- Rotate 45° about the origin and translate 1 unit in x and y direction
- Translate 1 unit in x and y direction and rotate 45° about the origin

Rotation

$$R = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R.T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 1 & 1 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A'(x', y') = (\frac{1}{\sqrt{2}} + 1, \frac{1}{\sqrt{2}} + 1)$$

$$B'(x', y') = (-\frac{1}{\sqrt{2}} + 1, \frac{1}{\sqrt{2}} + 1)$$

$$C'(x', y') = (1, 1)$$

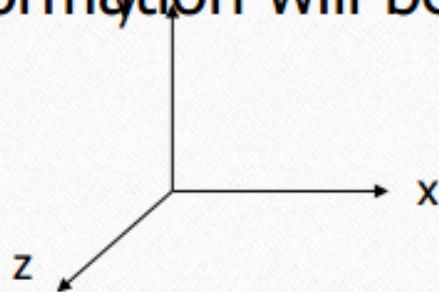
$$\begin{bmatrix} A'(x', y') \\ B'(x', y') \\ C'(x', y') \end{bmatrix} = \begin{bmatrix} x & y & w \\ Ax' & Ay' & 1 \\ Bx' & By' & 1 \\ Cx' & Cy' & 1 \end{bmatrix} = \begin{bmatrix} x & y & w \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} + 1 & \frac{1}{\sqrt{2}} + 1 & 0 \\ -\frac{1}{\sqrt{2}} + 1 & \frac{1}{\sqrt{2}} + 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Why use 3x3 matrices?

- So that we can perform all transformations using matrix/vector multiplications
- This allows us to *pre-multiply* all the matrices together
- The point (x,y) needs to be represented as $(x,y,1)$ -> this is called **Homogeneous coordinates!**

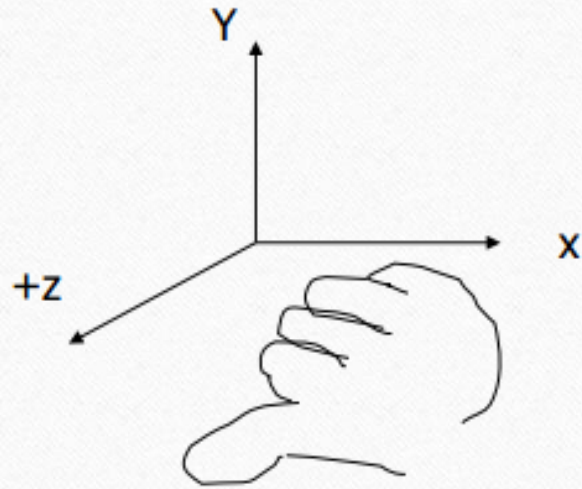
Three-Dimensional Graphics

- A 3D point (x,y,z) – x,y , and z coordinates
- We will still use column vectors to represent points
- Homogeneous coordinates of a 3D point $(x,y,z,1)$
- Transformation will be performed using 4x4 matrix

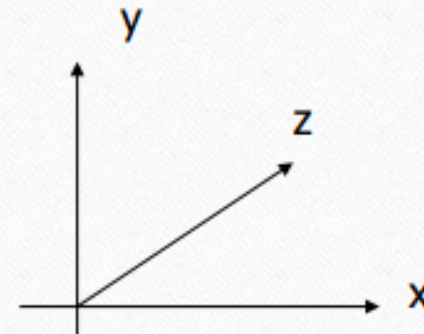


Right hand coordinate system

- $X \times Y = Z$; $Y \times Z = X$; $Z \times X = Y$;



Right hand coordinate system



Left hand coordinate system
Not used in this class and
Not in OpenGL

3D transformation

- Very similar to 2D transformation
- Translation

$$x' = x + tx; y' = y + ty; z' = z + tz$$

$$\begin{vmatrix} X' \\ Y' \\ Z' \\ 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & tx \\ 0 & 1 & 0 & ty \\ 0 & 0 & 1 & tz \\ 0 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} X \\ Y \\ Z \\ 1 \end{vmatrix}$$

homogeneous coordinates

3D transformation

- Scaling

$$X' = X * S_x; Y' = Y * S_y; Z' = Z * S_z$$

$$\begin{vmatrix} X' \\ Y' \\ Z' \\ 1 \end{vmatrix} = \begin{vmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} X \\ Y \\ Z \\ 1 \end{vmatrix}$$

3D transformation

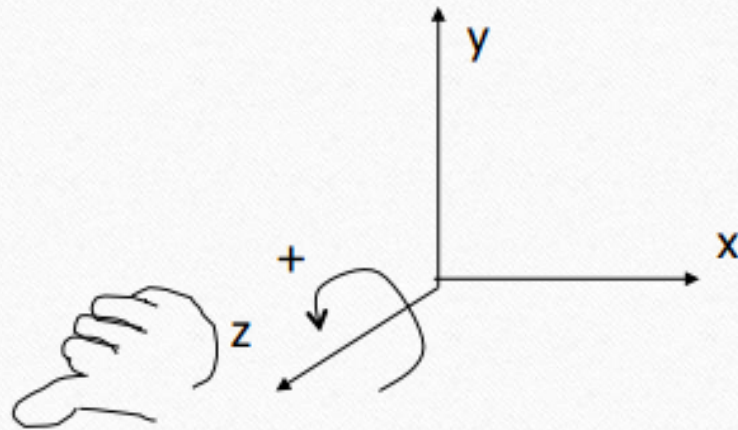
- Rotation about Z – similar to 2D rotation

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = x \sin(\theta) + y \cos(\theta)$$

$$z' = z$$

$$\begin{vmatrix} \cos(\theta) & -\sin(\theta) & 0 & 0 \\ \sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$



- OpenGL - `glRotatef(θ, 0,0,1)`

3D transformation

- Rotation about y ($z \rightarrow y, y \rightarrow x, x \rightarrow z$)

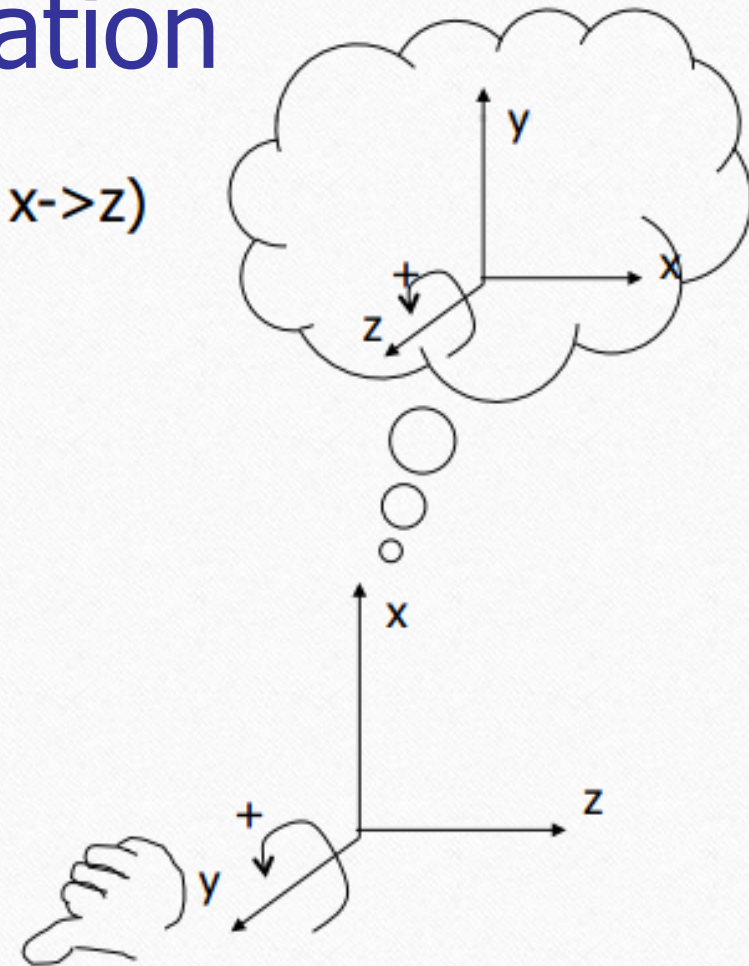
$$z' = z \cos(\theta) - x \sin(\theta)$$

$$x' = z \sin(\theta) + x \cos(\theta)$$

$$y' = y$$

$$\begin{vmatrix} \cos(\theta) & 0 & \sin(\theta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

- OpenGL - `glRotatef( $\theta$ , 0,1,0)`



3D transformation

- Rotation about x ($z \rightarrow x, y \rightarrow z, x \rightarrow y$)

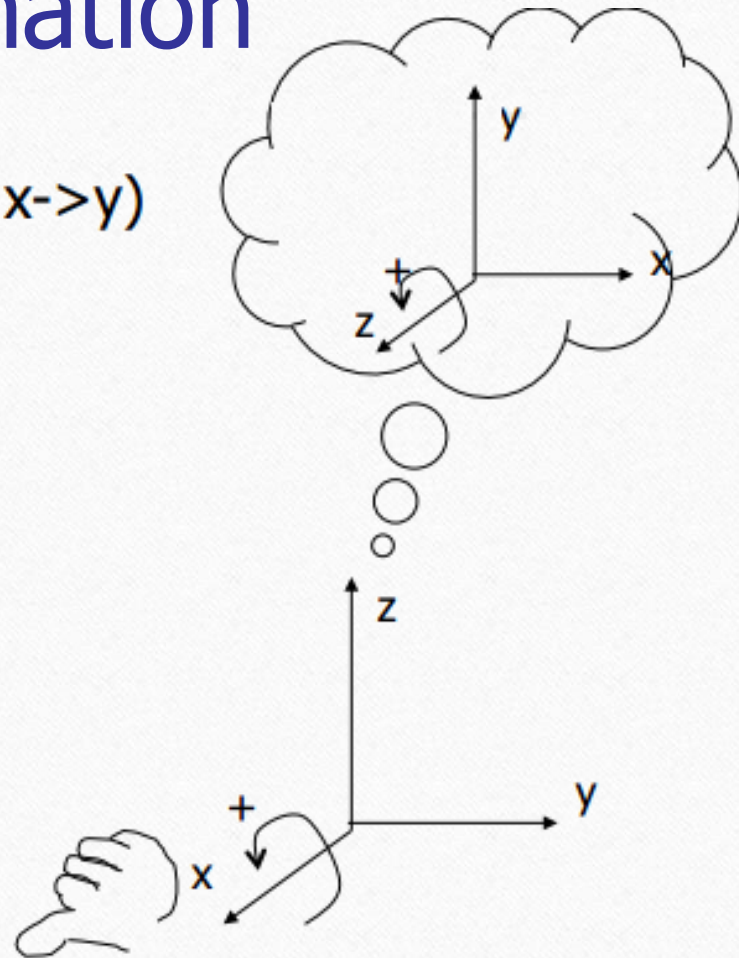
$$y' = y \cos(\theta) - z \sin(\theta)$$

$$z' = y \sin(\theta) + z \cos(\theta)$$

$$x' = x$$

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) & 0 \\ 0 & \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

- OpenGL - `glRotatef( $\theta$ , 1,0,0)`





Thank You